S M Zubeen

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SKILLS

Programming Languages: Python, Java, Git

Machine Learning & AI: Machine Learning, Deep Learning, Natural Language Processing, Generative AI, LLMs

Data Management: Data Science, Data cleaning and preprocessing, SQL

Computer Vision: OpenCV, Tensorflow, Mediapipe, EasyOCR, Pyteseract

Frameworks & Tools: TensorFlow, Keras, LangChain, Pandas, Numpy, NLTK, Pytorch, Mediapipe, Matplotlib

Other: FL Studio, Premier Pro

EXPERIENCE

Amazon ML Challenge:

Text Extraction and Entity Recognition in Images

- Developed and improved a text extraction pipeline for over 200,000 photos, increasing accuracy over Tesseract by 15% for complicated typefaces and noisy backgrounds.
- Integrated EasyOCR with BERT models, enhancing entity recognition by 20%.
- Troubleshooted model performance concerns with team members, coordinating and implementing fixes effectively through excellent communication.

PROJECTS

Drowsiness Detection:

Feb 2023 - Mar 2023

- Created a drowsiness detection tool that reduces training time by 30% by utilising pre-trained models. Annotated of over 1,000 photos in the dataset, enhancing team productivity and raising model accuracy by 10%.
- Employed systematic troubleshooting methods to rectify model errors in YOLOv5, achieving an impressive accuracy of 87%, which outperformed the prior RCNN results by a significant margin of 15%.

Face Mask Detection:

Nov 2022 - Dec 2022

- Developed a real-time face mask detection system using Python, OpenCV, and TensorFlow.
- Trained a CNN model with public datasets for mask/no mask classification, achieving 90% accuracy. Integrated model with OpenCV for real-time detection via webcam and Optimized detection speed for real-time performance.

Indian Sign Language Analysis:

May 2023 - Jun 2023

- Gesture recognition through the use of Mediapipe, TensorFlow, and OpenCV to refine object detection models. Achieved 82% accuracy by iteratively refining the model based on feedback and test results.

Video Segmentation and Impainting:

Jul 2024 - Present

- Implemented SAM for video object segmentation in coordination with partners, with an accuracy of 85% in recognition.
- Currently leveraging problem-solving skills to investigate Latent Diffusion models like VideoLDM to increase frame coherence and object replacement by 20%.

EDUCATION

Guru Tegh Bahadur Institute of Technology — **Rajouri Garden, New Delhi**
B. tech(AI-ML) — *Graduation Date: Jul 2025*

Arunodaya Public School — **Karkardooma, New Delhi**
Senior Secondary — *Graduation Date:Jul 2021*